

Security in the age of AI

We speak to Juniper Networks India to see how security agencies are evolving to meet the growing threat of AI

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How are security threats evolving with the ushering of disruptive technologies such as AI/ML?

Sriram: Threat sophistication over the last few years are increasingly challenging to identify and mitigate. Adaptive malware sails through undetected across traditional security solutions, posing and behaving like genuine traffic.

AI/ML technologies are being used for launching persistent and targeted attacks including zero-day attacks, that gives very little time to identify threats and mitigate the same. It's becoming a battle against machines and automation.

Cybercrime is also becoming more and more organized – more than 76% of breaches are financially motivated. Because of this, the price of fighting cybercrime has inevitably become more exorbitant, with an estimated USD\$8 trillion to be spent globally on fighting cybercrime by 2022.

How is Juniper leveraging AI/ML to protect businesses from data breaches?

Sriram: Juniper leverages AI/ML for behaviour analysis of the traffic/user and building the threat intelligence based on data collected across multiple sources over a period of time. This results in faster detection of advanced threats, behaviour analytics and one-step mitiga-

tion across web, email and lateral (east-west) traffic.

With Juniper's Contrail Security, enterprises and service providers can now further optimize their multi-cloud environments, while mitigating security risks by enabling application traffic discovery through automated policy enforcement.

How can CISOs be better prepared for a cyber-attack?

Sriram: CISOs should help organizations build a stronger security posture to achieve adaptive and immediate protection through automated and actionable intelligence; one that transforms the network into a single and holistic defence domain, wherein every element becomes an enforcement point. When building a strong security architecture, it's important to keep the following in mind:

- Utilization of scarce security personnel – streamlined incident investigation and one-touch remediation
- Simplicity driven – unified management and visibility
- Deployment flexibility – on prem, virtual, cloud, or blended with open, robust architecture
- Investment protection – more efficient operations with simplified architecture

Why is IoT security different, and what new approaches are required?

Sriram: Proliferation of IoT devices and

distributed mobile edge computing (MEC) are two critical aspects of IoT landscape. IoT device count is expected to reach 31 billion by 2020 and this increases the attack surface significantly. Distributed IoT computing will expand considerably to cater to low-latency applications and hence will result in several vulnerable computing nodes in the IoT landscape. Hence, security in IoT domain has to be addressed and approached accordingly with the following in mind:

- Securing IoT – compute everywhere, security everywhere
- Protect the edge – devices, gateways, network connection and applications
- Protect the network – every network element is a source of intelligence, and is an enforcement point
- Protection in the cloud – data center, cloud-based analytics, machine learning, policy framework
- V2X networks (vehicle to anything) – V2V (vehicle), V2P (person), V2N (network), V2I (infra)
- With distributed computing comes distributed security. IoT gateways are also becoming distributed data centers – it's vital to increase gateway, network and app protection, whether on the edge (physical/virtual), or on the cloud or anywhere (hybrid).
- IoT security should also be battery efficient to ensure low throughput connectivity. 



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